

Spring-Energy Clustering Is Kernel Clustering:

An Equivalence Theorem Behind a Null Screening

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Abstract

A research line on “mechanical analogies for data clustering” screened clustering objectives built from springs, tension and lifting against matched kernel controls and found no mechanical component that survived the comparison. No manuscript or code of that screening was available to us; Section 2 reconstructs the family from the line’s public summary as *pairwise* spring energies (points joined by springs with an arbitrary pair potential $\phi(d)$, optionally after a lift, with the within-cluster energy normalised per particle or summed), and this note shows that, for that pairwise family, the null result is a theorem. We prove that (i) Hooke springs with zero rest length give exactly the k -means objective; (ii) for every pair potential the per-particle energy is, up to an additive constant and a factor $\frac{1}{2}$, the kernel k -means objective of $-\phi(D)$, which a diagonal shift makes positive semidefinite without changing the minimisers (the ratio-association form of kernel k -means of Dhillon, Guan and Kulis, written for spring matrices); (iii) when $\phi(d) = \psi(d^2)$ with ψ a Bernstein function, Schoenberg’s theory supplies a positive definite kernel that does not depend on the dataset once the scales are fixed, so the spring energy is, up to an additive constant, the Hooke energy of a lifted configuration (feature map rescaled by $1/\sqrt{2}$); (iv) lifting changes only the kernel; (v) the total energy and its “tension” dual are min-sum k -clustering and max- k -cut with weights $\phi(d_{ij})$; and (vi) the zero-temperature relaxation of the spring system is Hartigan’s single-move method, whose fixed points are always Lloyd fixed points of the kernel problem. Exact rational arithmetic on two explicit configurations shows that the three-body (triangle-area) term, the cluster-mean rest length and the per-spring average tested here are not pairwise objectives. Computationally, on five datasets and six potentials the spring energies and the kernel objectives coincide to 1.1×10^{-11} on 6090 partitions, a brute-force spring implementation and a kernel implementation of the same relaxation follow identical trajectories in 90/90 runs, and all 600 relaxation fixed points are kernel k -means fixed points. Dynamics on positions rather than assignments is the one direction the theorem does not close; we state it as an open problem. No claim is made about clustering performance.

1 Introduction

Physical metaphors recur in clustering: algorithms have been built on gravity [29], on spin models at finite temperature [3], on deterministic annealing of a free energy [19, 20], on quantum potentials [14], and on elastic nets [9, 30]. The research line to which this note belongs (LMI001, “Mechanical analogies for data clustering”) studied formulations of a more elementary kind: points connected by springs, with tension and lifting terms, clustered by minimising an elastic energy. Its stated objective was to determine whether the mechanical component produces an effect distinguishable from matched kernel controls, and its screening found none. *The definitions of Section 2 are a reconstruction from the line’s public summary; no manuscript or code of the original screening was available to us, and the conclusions below apply to objectives of that pairwise form.* The purpose of the note is to explain the null result rather than to repeat it. The

explanation is short: every objective in the pairwise family is a kernel objective, and the natural way to minimise it mechanically is a known kernel k -means optimiser. A control that matches the kernel therefore matches the objective exactly, and no experiment on objective values can separate them (experiments on optimisers can, Section 5). We proceed in three steps: we define the family (Section 2); we prove the equivalences (Section 3), whose core is elementary linear algebra around the trace form of kernel k -means and its ratio-association reading [6, 8, 11, 31], the only non-trivial input being Schoenberg’s characterisation of metric transforms of Hilbertian type [2, 23, 24], and we certify by exact arithmetic that the ingredients the theorem does not cover are genuinely outside the family (Section 4); and we confirm the identities computationally with predefined criteria (Section 5).

Scope. The note is about objectives and their minimisers, not about clustering quality. We do not claim that any potential clusters real data well, and the adjusted Rand indices reported are context, not results. The datasets are standard synthetic ones and two small scikit-learn datasets [18]. Every number in the text is generated by the scripts described in Appendix A, with fixed seeds. Section 6 lists all claims with their status.

2 The pairwise spring family

Throughout, $X = (x_1, \dots, x_n)$ is a configuration in $\mathbb{R}^d$, $k \geq 2$ is fixed, and a *partition* $C = (C_1, \dots, C_k)$ is a partition of $[n]$ into k non-empty clusters, with sizes $n_c = |C_c|$ and indicator vectors $\mathbf{1}_c \in \{0, 1\}^n$. We write $D_{ij} = \|x_i - x_j\|$. “PSD” always refers to a matrix (positive semidefinite); a *kernel* on a set is called *positive definite* (PD) when all its Gram matrices are PSD, the convention of Berg et al. [2].

Definition 2.1 (Spring network, elastic energy, variants). A *spring network* is a triple (ϕ, L, w) of a *pair potential* $\phi: [0, \infty) \rightarrow \mathbb{R}$, a *lift* $L: \mathbb{R}^d \rightarrow \mathbb{R}^D$ (the identity by default), and a *normalisation* $w \in \{w_{\text{sp}}, w_{\text{tot}}\}$ with $w_{\text{sp}}(m) = 1/m$ (energy per particle of the cluster) and $w_{\text{tot}}(m) = 1$ (total energy). Its *within-cluster elastic energy* is

$$E_{\phi, L, w}(C) = \sum_{c=1}^k w(n_c) \sum_{\substack{i < j \\ i, j \in C_c}} \phi(\|Lx_i - Lx_j\|). \quad (1)$$

We write $A_{ij} = \phi(\|Lx_i - Lx_j\|)$ for the *spring matrix*, E_{sp} and E_{tot} for the two normalisations, and $\text{SSE}(C) = \sum_c \sum_{i \in C_c} \|x_i - \mu_c\|^2$, μ_c the centroid, for the k -means objective [16, 17]. The *released tension* of C is the energy of the springs that cross clusters, $T(C) = \sum_{c < c'} \sum_{i \in C_c, j \in C_{c'}} A_{ij}$; a tension formulation maximises T . An *anchored* network attaches every particle to a hub $y_c \in \mathbb{R}^D$ of its cluster, with energy $E^{\text{anch}}(C, y) = \sum_c \sum_{i \in C_c} \phi(\|Lx_i - y_c\|)$, minimised over the hubs as well as over C .

The potentials used below are: the Hooke spring with zero rest length, $\phi(d) = d^2$ (the spring constant is absorbed); the Hooke spring with rest length $r > 0$, $\phi(d) = (d - r)^2$; the string under constant tension, $\phi(d) = d$; the saturating (breakable) spring, $\phi(d) = 1 - e^{-d^2/2\ell^2}$; the soft spring, $\phi(d) = \log(1 + d^2/\ell^2)$; and the Hooke spring on the paraboloid lift $L(x) = (x, \|x\|^2/s)$.

Definition 2.2 (Kernel k -means objective, trace form). For a symmetric matrix $K \in \mathbb{R}^{n \times n}$ and a partition C ,

$$J_K(C) = \text{tr } K - \sum_{c=1}^k \frac{\mathbf{1}_c^\top K \mathbf{1}_c}{n_c}. \quad (2)$$

When K is PSD with a Gram factorisation $K_{ij} = \langle \Phi_i, \Phi_j \rangle$, $J_K(C) = \sum_c \sum_{i \in C_c} \|\Phi_i - m_c\|^2$ with m_c the centroid of the Φ_i , $i \in C_c$: the k -means objective in feature space [6, 11, 25, 31]. For general symmetric K we take (2) as the definition.

Definition 2.3 (Negative type, Bernstein functions). A symmetric kernel ψ on a set S is *conditionally negative definite* (CND) if $\sum_{i,j} c_i c_j \psi(s_i, s_j) \leq 0$ for all finite families $s_i \in S$ and reals c_i with $\sum_i c_i = 0$. A function $\phi: [0, \infty) \rightarrow \mathbb{R}$ is of *negative type* if $(x, y) \mapsto \phi(\|x - y\|)$ is CND on $\mathbb{R}^d$ for every d . A function $\psi: [0, \infty) \rightarrow [0, \infty)$ is a *Bernstein function* if it is continuous, C^∞ on $(0, \infty)$, and $(-1)^{m-1} \psi^{(m)} \geq 0$ for all $m \geq 1$; equivalently (Lévy–Khintchine representation, [22, Thm. 3.2]) $\psi(s) = a + bs + \int_{(0,\infty)} (1 - e^{-ts}) \mu(dt)$ with $a, b \geq 0$ and $\int \min(1, t) \mu(dt) < \infty$.

3 Results

3.1 Invariances of the trace form

Lemma 3.1 (What J does not see). *Let K be symmetric, C a partition into k non-empty clusters, $\sigma, a \in \mathbb{R}$ and $f \in \mathbb{R}^n$. Then*

$$J_{K+\sigma I}(C) = J_K(C) + \sigma(n - k), \quad J_{K+a\mathbf{1}\mathbf{1}^\top}(C) = J_K(C), \quad J_{K+f\mathbf{1}^\top + \mathbf{1}f^\top}(C) = J_K(C).$$

Proof. $\text{tr}(\sigma I) = \sigma n$ and $\mathbf{1}_c^\top \sigma I \mathbf{1}_c / n_c = \sigma$ for each of the k clusters. $\text{tr}(a\mathbf{1}\mathbf{1}^\top) = an$ and $\sum_c an_c^2 / n_c = an$. $\text{tr}(f\mathbf{1}^\top + \mathbf{1}f^\top) = 2\sum_i f_i$ and $\mathbf{1}_c^\top (f\mathbf{1}^\top + \mathbf{1}f^\top) \mathbf{1}_c / n_c = 2\sum_{i \in C_c} f_i$, which sums to $2\sum_i f_i$ over c . $\square$

In particular the minimisers of J_K over partitions into k clusters are unchanged by a diagonal shift, by a constant added to every entry, and by a separable term $f_i + f_j$. The diagonal shift is the device of Dhillon et al. [6, 8] for making an indefinite affinity matrix PSD without changing the optimal partitions; the other two invariances are what connects the spring energy to Schoenberg’s theory below.

3.2 Springs are kernels

Theorem 3.2 (Spring-energy clustering is kernel clustering). *Let (ϕ, L, w) be a spring network with spring matrix A and let $K = -A$ (diagonal included, $K_{ii} = -\phi(0)$). For every partition C into k non-empty clusters:*

- (a) $E_{\phi, L, w}(C)$ depends on the data only through A .
- (b) (per-particle normalisation) $E_{\text{sp}}(C) = \frac{1}{2}J_K(C) + \frac{1}{2}(n - k)\phi(0)$. For every $\sigma \geq -\lambda_{\min}(K)$ the matrix $K + \sigma I$ is PSD and $E_{\text{sp}}(C) = \frac{1}{2}J_{K+\sigma I}(C) + \frac{1}{2}(n - k)(\phi(0) - \sigma)$; hence $\arg \min_C E_{\text{sp}} = \arg \min_C J_{K+\sigma I}$, the k -means problem in the feature space of any Gram factorisation of $K + \sigma I$.
- (c) (Hooke) If $\phi(d) = d^2$ and L is the identity, $E_{\text{sp}}(C) = \text{SSE}(C)$ exactly.
- (d) (total normalisation and tension) $E_{\text{tot}}(C) + T(C) = \sum_{i < j} A_{ij}$ is independent of C ; minimising E_{tot} is maximising the released tension, which is the max- k -cut, equivalently min-sum k -clustering, problem with weights A_{ij} [21]. If all clusters have the same size, $E_{\text{tot}} = (n/k)E_{\text{sp}}$.
- (e) (converse) Every kernel k -means objective is a Hooke energy: if K is PSD and $K = \Phi\Phi^\top$, then $J_K(C) = E_{d^2, \Phi, \text{sp}}(C)$, the per-particle Hooke energy of the lifted configuration $(\Phi_1, \dots, \Phi_n)$.
- (f) (universal kernel) If ϕ is of negative type, then

$$\widetilde{K}(x, y) = \phi(\|x\|) + \phi(\|y\|) - \phi(\|x - y\|) - \phi(0)$$

is a PD kernel on $\mathbb{R}^D$ that does not depend on the dataset (given the scales ℓ, r, s that enter ϕ and L), and $E_{\text{sp}}(C) = \frac{1}{2}J_{\widetilde{K}}(C) + \frac{1}{2}(n-k)\phi(0)$ where $\widetilde{K}$ is evaluated on the lifted points. If Φ is a feature map of $\widetilde{K}$ and $\widetilde{\Phi} = \Phi/\sqrt{2}$, then $\|\widetilde{\Phi}(x) - \widetilde{\Phi}(y)\|^2 = \phi(\|x - y\|) - \phi(0)$ and

$$E_{\text{sp}}(C) = E_{d^2, \widetilde{\Phi} \circ L, \text{sp}}(C) + \frac{1}{2}(n-k)\phi(0) :$$

up to the constant, the spring energy is the per-particle Hooke energy of the lifted configuration $(\widetilde{\Phi}(Lx_i))_i$. (Without the rescaling, E_{sp} is half the Hooke energy of $(\Phi(Lx_i))_i$ plus the constant.)

Status. (a), (d) elementary; (c) classical. (b) is the ratio-association form of kernel k -means: $\sum_c \mathbf{1}_c^\top A \mathbf{1}_c / n_c$ is the ratio association of the affinity A , whose equivalence with kernel k -means for $K = \sigma I \mp A$ is due to Dhillon et al. [6, 8]; specific to springs is only $E_{\text{sp}} = \frac{1}{2}(\text{ratio association}) - \frac{1}{2}k\phi(0)$ with the explicit constant [literature]. (e) is the feature-space reading of k -means [11, 31] [classical]. (f) [proved here from the CND-PD lemma of 2].

Proof. (a) is the definition. (b) Since $\sum_{i < j \in C_c} A_{ij} = \frac{1}{2}(\mathbf{1}_c^\top A \mathbf{1}_c - n_c \phi(0))$,

$$E_{\text{sp}}(C) = \frac{1}{2} \sum_c \frac{\mathbf{1}_c^\top A \mathbf{1}_c}{n_c} - \frac{1}{2}k\phi(0), \quad \text{and} \quad \sum_c \frac{\mathbf{1}_c^\top A \mathbf{1}_c}{n_c} = - \sum_c \frac{\mathbf{1}_c^\top K \mathbf{1}_c}{n_c} = J_K(C) - \text{tr } K,$$

with $\text{tr } K = -n\phi(0)$, so $E_{\text{sp}}(C) = \frac{1}{2}J_K(C) + \frac{1}{2}(n-k)\phi(0)$. The shifted form follows from Lemma 3.1, and $K + \sigma I \succeq 0$ for $\sigma \geq -\lambda_{\min}(K)$. (c) For any finite set P with centroid μ , $\sum_{i,j \in P} \|x_i - x_j\|^2 = 2|P| \sum_{i \in P} \|x_i - \mu\|^2$ (expand $\|(x_i - \mu) - (x_j - \mu)\|^2$; the cross terms vanish). Halving and dividing by $|P|$ gives $\frac{1}{n_c} \sum_{i < j \in C_c} d_{ij}^2 = \sum_{i \in C_c} \|x_i - \mu_c\|^2$; sum over c . (d) Every pair $\{i, j\}$ is either within a cluster or across, so $E_{\text{tot}} + T$ is the total spring energy; the equal-size statement is $w_{\text{tot}} = (n/k)u_{\text{sp}}$. (e) By (b) with $\phi(0) = 0$ applied to the configuration Φ : $E_{d^2, \Phi, \text{sp}}(C) = \frac{1}{2}J_{-D_\Phi^{(2)}}(C)$, where $(D_\Phi^{(2)})_{ij} = K_{ii} + K_{jj} - 2K_{ij}$, so $-D_\Phi^{(2)} = 2K - f\mathbf{1}^\top - \mathbf{1}f^\top$ with $f_i = K_{ii}$, and Lemma 3.1 gives $\frac{1}{2}J_{2K}(C) = J_K(C)$. (f) Positive definiteness of $\widetilde{K}$ is the lemma relating CND and PD kernels [2, Ch. 3, §2, Lemma 2.1], whose proof we include: given points $x_1, \dots, x_m$, reals $c_1, \dots, c_m$, set $x_0 = 0$ and $c_0 = -\sum_{i \geq 1} c_i$; the CND inequality for the family $(x_i, c_i)_{i \geq 0}$ reads $\sum_{i,j \geq 1} c_i c_j \phi(\|x_i - x_j\|) + 2c_0 \sum_{i \geq 1} c_i \phi(\|x_i\|) + c_0^2 \phi(0) \leq 0$, and substituting c_0 this is $\sum_{i,j \geq 1} c_i c_j [\phi(\|x_i - x_j\|) - \phi(\|x_i\|) - \phi(\|x_j\|) + \phi(0)] \leq 0$, i.e. $\sum_{i,j} c_i c_j \widetilde{K}(x_i, x_j) \geq 0$. On the dataset, $\widetilde{K} = K + f\mathbf{1}^\top + \mathbf{1}f^\top - \phi(0)\mathbf{1}\mathbf{1}^\top$ with $f_i = \phi(\|Lx_i\|)$, so $J_{\widetilde{K}} = J_K$ by Lemma 3.1 and the identity follows from (b). Next, $\|\Phi(x) - \Phi(y)\|^2 = \widetilde{K}(x, x) + \widetilde{K}(y, y) - 2\widetilde{K}(x, y) = 2\phi(\|x - y\|) - 2\phi(0)$, so $\|\widetilde{\Phi}(x) - \widetilde{\Phi}(y)\|^2 = \phi(\|x - y\|) - \phi(0)$. Finally (e) applied to $\widetilde{K}$ gives $J_{\widetilde{K}}(C) = E_{d^2, \Phi \circ L, \text{sp}}(C)$, and scaling a configuration by $1/\sqrt{2}$ halves every squared distance, so $\frac{1}{2}J_{\widetilde{K}}(C) = E_{d^2, \widetilde{\Phi} \circ L, \text{sp}}(C)$. For Hooke springs $\widetilde{K}(x, y) = 2\langle x, y \rangle$, $\Phi(x) = \sqrt{2}x$ and $\widetilde{\Phi}(x) = x$. $\square$

Proposition 3.3 (Which springs are of negative type). (i) $\phi(d) = d^2$ is of negative type: for $\sum_i c_i = 0$, $\sum_{i,j} c_i c_j \|x_i - x_j\|^2 = -2\|\sum_i c_i x_i\|^2 \leq 0$. [classical]

(ii) If N is a CND kernel with $N \geq 0$ and ψ is a Bernstein function, then $\psi \circ N$ is CND. [literature]

(iii) Consequently $\phi(d) = \psi(d^2)$ is of negative type for every Bernstein ψ ; this covers $d^{2\alpha}$ for $0 < \alpha \leq 1$ (in particular d), $\log(1 + d^2/\ell^2)$, $1 - e^{-d^2/2\ell^2}$ and $d^2/(1 + d^2/\ell^2)$. The rest-length potential $(d - r)^2 = d^2 - 2rd + r^2$ is a difference of two functions of negative type and is not covered.

Proof sketch. (i) Expand $\|x_i - x_j\|^2 = \|x_i\|^2 + \|x_j\|^2 - 2\langle x_i, x_j \rangle$; the first two terms vanish against $\sum c_i = 0$. (ii) is proved in Berg et al. [2, Ch. 3, §2] (Schoenberg's theorem is Theorem 2.2 there;

the composition with Bernstein functions follows from it and the representation in Definition 2.3), see also Schilling et al. [22]; we do not vouch for the exact numbering of the composition result. The argument is: by Schoenberg’s theorem [24] N is CND iff e^{-tN} is PD for all $t > 0$, so $1 - e^{-tN}$ is CND (a constant kernel is CND-neutral and the negative of a PD kernel is CND); write $\psi(N) = a + bN + \int (1 - e^{-tN})\mu(dt)$ and note that non-negative combinations and pointwise limits of CND kernels are CND. (iii) The listed ψ are Bernstein: s^α ($0 < \alpha \leq 1$), $\log(1 + s/\ell^2)$, $1 - e^{-s/2\ell^2}$, $s/(1 + s/\ell^2)$ all have completely monotone derivatives. *Rest length.* Theorem 3.2(b) still applies to $(d - r)^2$: its spring energy is the kernel k -means objective of $-(D - r)^{(2)}$ after a diagonal shift, and its minimisers are feature-space k -means minimisers for that dataset. What (iii) withholds is a feature map defined on all of $\mathbb{R}^d$; E1 below finds the kernel $\widetilde{K}$ indefinite for this potential on every dataset (Table 1), consistent with the expectation, although we do not prove that no dataset-independent PSD representative exists. $\square$

Remark 3.4 (Lifting and anchoring). (a) $E_{\phi,L,w}(X) = E_{\phi,\text{id},w}(LX)$, so a lift replaces A by $\phi(\|Lx_i - Lx_j\|)$ and nothing else; for Hooke springs a lift is exactly the choice of the kernel $\langle Lx, Ly \rangle$, and by Theorem 3.2(e) every PSD kernel arises this way [trivial]. (b) For Hooke springs, $\min_y E^{\text{anch}}(C, y) = E_{\text{sp}}(C) = \text{SSE}(C)$, attained at the centroids (Theorem 3.2(c) and the fact that the centroid minimises the sum of squared distances), and the alternating equilibration “hubs move to the equilibrium of their springs; particles re-attach to the hub of least energy” is Lloyd’s algorithm [16] by definition. For a general ϕ , $\min_y \sum_{i \in C_c} \phi(\|x_i - y\|)$ is a Weber-type location problem; with $\phi(d) = d$ the anchored objective is the k -median objective (geometric medians as hubs) [classical]. We make no claim about whether these anchored objectives are of the form (1).

3.3 Relaxation of the spring system

The mechanical way to minimise (1) is to let the system relax: one particle at a time jumps to the cluster in which its springs store less energy, or, at finite temperature, jumps with Metropolis probability. On assignments this is a known algorithm.

Proposition 3.5 (Zero-temperature relaxation is Hartigan’s method). *Let K' be any PSD matrix with $J_{K'} = J_K + \text{const}$ on partitions (for instance $K + \sigma I$ or, for ϕ of negative type, $\widetilde{K}$), and write $\delta_{K'}(i, c) = K'_{ii} - \frac{2}{n_c}(K'\mathbf{1}_c)_i + \frac{1}{n_c^2}\mathbf{1}_c^\top K'\mathbf{1}_c$ for the squared feature distance from Φ_i to the centroid of cluster c . Moving particle i from its cluster a ($n_a \geq 2$) to cluster c changes E_{sp} by*

$$\Delta E = \frac{1}{2} \left[\frac{n_c}{n_c+1} \delta_{K'}(i, c) - \frac{n_a}{n_a-1} \delta_{K'}(i, a) \right], \quad (3)$$

and the right-hand side does not depend on the representative K' . Consequently:

- (i) the greedy single-particle relaxation of the spring energy is Hartigan’s method [12, 13] applied to the kernel k -means objective;
- (ii) every partition that is stable under single-particle moves is Voronoi-stable in the feature space of every PSD representative K' , i.e. each Φ_i is assigned to a nearest centroid, so it is a fixed point of Lloyd’s algorithm for K' with ties broken in favour of the current cluster (on the relation between such fixed points and local optimality see 26);
- (iii) the converse fails: $X = \{0, 2, 3.4, 3.6\} \subset \mathbb{R}$, $k = 2$, $C = \{\{0, 2\}, \{3.4, 3.6\}\}$ is Voronoi-stable with $\text{SSE} = 2.02$, while moving the point 2 gives $\text{SSE} = 1.52$.

Proof. Write $S_{ic} = (K'\mathbf{1}_c)_i$ and $Q_c = \mathbf{1}_c^\top K'\mathbf{1}_c$. Removing i from a changes Q_a/n_a to $(Q_a - 2S_{ia} + K'_{ii})/(n_a - 1)$, adding it to c changes Q_c/n_c to $(Q_c + 2S_{ic} + K'_{ii})/(n_c + 1)$, and $\Delta J = -\Delta \sum_c Q_c/n_c$. Expanding both differences and comparing with $\delta_{K'}$ gives $\Delta J = \frac{n_c}{n_c+1} \delta_{K'}(i, c) - \frac{n_a}{n_a-1} \delta_{K'}(i, a)$, an algebraic identity valid for any symmetric K' ; $\Delta E = \frac{1}{2} \Delta J$ by Theorem 3.2(b). Independence

of the representative: if $J_{K'} = J_K + \text{const}$ on partitions then $\Delta J_{K'} = \Delta J_K$ for every move, and the identity holds for both matrices, so the right-hand side of (3) is the same for every such K' . Explicitly, under $K' \mapsto K' + \sigma I$, $\delta(i, c)$ increases by $\sigma(1 + 1/n_c)$ for $i \notin C_c$ and by $\sigma(1 - 1/n_a)$ for $i \in C_a$, and both terms of (3) increase by σ ; under the other two invariances of Lemma 3.1, δ itself is unchanged. (i) is the definition of Hartigan’s move. (ii) If no move decreases E then $\frac{n_c}{n_c+1}\delta(i, c) \geq \frac{n_a}{n_a-1}\delta(i, a)$ for all $c \neq a$, and since $\frac{n_c}{n_c+1} < 1 < \frac{n_a}{n_a-1}$ this forces $\delta(i, c) \geq \delta(i, a)$ (not necessarily strictly, whence the tie rule). (iii) Centroids 1 and 3.5; the point 2 is at distance 1 from the first and 1.5 from the second; after the move the clusters $\{0\}$ and $\{2, 3.4, 3.6\}$ (centroid 3) have $\text{SSE} = 0 + 1 + 0.16 + 0.36 = 1.52$; by (3) with $K' = \langle x, y \rangle$, $\Delta \text{SSE} = \frac{2}{3} \cdot 1.5^2 - 2 \cdot 1^2 = -0.5$. $\square$

Item (ii) generalises the observation of Telgarsky and Vattani [27] for Euclidean k -means; item (iii) is the standard type of example. The consequence for the research line is that “spring relaxation” cannot converge to anything outside the set of kernel k -means fixed points.

Proposition 3.6 (Lloyd’s rule is representation-dependent). *For $K' = K + \sigma I$, $\delta_{K'}(i, c) = \delta_K(i, c) + \sigma(1 + \frac{1-2 \cdot [i \in C_c]}{n_c})$: the shift adds $\sigma(1 + 1/n_c)$ to the squared distance from i to every other cluster c and only $\sigma(1 - 1/n_a)$ to the distance to its own cluster a . Hence the Voronoi rule “assign i to $\arg \min_c \delta_{K'}(i, c)$ ” favours staying, and as $\sigma \rightarrow \infty$ every partition becomes Voronoi-stable, while E_{sp} and the single-move dynamics are unchanged. [proved here; explicit form of an effect noted by 7, 8]*

Proof. $(\sigma I \mathbf{1}_c)_i = \sigma[i \in C_c]$ and $\mathbf{1}_c^\top \sigma I \mathbf{1}_c = \sigma n_c$; substitute in δ . $\square$

Dhillon et al. [7, 8] already observe that a too large diagonal shift degrades the clusters that kernel k -means finds; the proposition is the explicit mechanism and E3 quantifies it. So “kernel k -means with Lloyd’s iteration” is not one algorithm for a spring energy but one per PSD representative; the single-move relaxation is the representative-free one. For springs of negative type the natural representative is $\widetilde{K}$ of Theorem 3.2(f), which for Hooke springs is $2\langle x, y \rangle$ and reproduces ordinary Lloyd k -means; the experiments use it, and use the naively shifted $-A + \sigma I$ only as a control (E3).

4 What leaves the family

Theorem 3.2 covers every objective that is a sum over within-cluster *pairs* of a fixed function of the data, weighted by a function of the cluster size in $\{1, 1/n_c\}$. An ingredient can escape only by breaking one of these three features: the sum over pairs (many-body terms), the fixedness of the pair function (parameters that depend on the assignment), or the normalisation. A fourth route replaces the minimisation over partitions by a dynamics on positions. For the first three, specific instances can be certified exactly.

Theorem 4.1 (Exact separation on explicit configurations; computer-assisted). *Let P_1 and P_2 be the configurations of 7 points in $\mathbb{Q}^2$*

$$\begin{aligned} P_1 &= \{(5, 10), (6, 2), (4, 4), (2, 2), (3, 7), (0, 6), (8, 8)\}, \\ P_2 &= \{(0, 2), (7, 4), (8, 7), (5, 0), (2, 8), (2, 3), (4, 10)\}, \end{aligned}$$

drawn once from a dedicated generator (seed 20260932), and let $k = 2$, so that each has 63 partitions. Call a partition function F w -pairwise if $F(C) = \sum_c w(n_c) \sum_{i < j \in C_c} A_{ij} + a$ for some symmetric A and constant a (22 unknowns). Then, with squared distances $s_{ij} = \|x_i - x_j\|^2$, the following holds on both configurations (relative residuals on $P_1; P_2$):

partition function	$w_{\text{sp}}\text{-pairwise?}$	$w_{\text{tot}}\text{-pairwise?}$
$\sum_c \frac{1}{n_c} \sum d_{ij}^2$ (Hooke, per particle)	yes (0; 0)	no (0.017; 0.015)
$\sum_c \sum d_{ij}^2$ (Hooke, total)	no (0.179; 0.179)	yes (0; 0)
$\sum_c \frac{1}{n_c} \sum (d_{ij}^2 - \rho)^2$, fixed ρ	yes (0; 0)	no (0.026; 0.030)
$\sum_c \frac{1}{n_c} \sum (d_{ij}^2 - \rho_c)^2$, $\rho_c = \text{cluster mean}$	no (0.143; 0.157)	no (0.109; 0.122)
$\sum_c \frac{1}{n_c} \sum_{i<j<l} 4 \text{area}(ijl)^2$ (three-body)	no (0.331; 0.328)	no (0.039; 0.037)
$\sum_c \binom{n_c}{2}^{-1} \sum d_{ij}^2$ (per spring)	no (0.185; 0.180)	no (0.125; 0.121)

where ρ is the mean of all s_{ij} , ρ_c the mean of s_{ij} over the pairs of cluster c , and $\text{area}(ijl)$ the area of the triangle $x_i x_j x_l$. The yes/no pattern agrees on the two configurations in 12 of the 12 cells.

Proof. For fixed w the w -pairwise functions form a linear subspace of $\mathbb{R}^{63}$ spanned by the columns of an explicit rational design matrix (rank 21 for w_{sp} , 22 for w_{tot} , on both configurations). Membership is decided by exact least squares in rational arithmetic (`fractions.Fraction`): a zero residual exhibits A and a ; a non-zero residual is a certificate that F is not in the span. The script `experiments/identity_check.py` (E4) performs the computation; the relative residuals are reported in the table. $\square$

The two positive controls (Hooke and fixed rest length under their own normalisation, total Hooke under w_{tot}) confirm that the test is not blind. The rest of the table says: the two normalisations are genuinely different objectives (each fails to be pairwise under the other’s weight), the per-spring average is neither, and the cluster-mean rest length and the triangle-area three-body term are outside the family in either normalisation. One instance suffices for each of these “not in general” statements, and the theorem makes exactly those: it certifies the three functionals tested, not the classes “many-body terms” or “assignment-dependent rest lengths”. The class statements would be false: for clusters of size $n_c \geq 3$, $\sum_{i<j<l \in C_c} (s_{ij} + s_{jl} + s_{il}) = (n_c - 2) \sum_{i<j \in C_c} s_{ij}$, so a three-body sum with a suitable size weight collapses to a pairwise energy.

The dynamic route. Several physics-inspired algorithms do not minimise an energy over partitions but move the *positions*: gravitational clustering [29], mean shift [4, 5, 10] (a gradient ascent on a kernel density estimate, hence a dynamics under a pairwise potential), quantum clustering [14], and the finite-temperature correlations of a Potts model [3]. Their output is a function of the trajectory, and Theorem 3.2 says nothing about it.

Open problem 4.2 (Dynamic mechanical clustering). Let $V: (0, \infty) \rightarrow \mathbb{R}$ be a pair potential, let $z(t)$ solve $\dot{z}_i = - \sum_{j \neq i} V'(\|z_i - z_j\|) \frac{z_i - z_j}{\|z_i - z_j\|}$ with $z(0) = X$, and let $P_{T,\varepsilon}(X)$ be the single-linkage partition of $z(T)$ at threshold ε . Characterise the triples (V, T, ε) for which there exist a fixed function $a: [0, \infty) \rightarrow \mathbb{R}$ and a normalisation w such that, for every X , $P_{T,\varepsilon}(X)$ minimises $\sum_c w(n_c) \sum_{i<j \in C_c} a(\|x_i - x_j\|)$ over partitions into the number of clusters it produces.

Conjecture 4.3. For the Gaussian attraction $V(d) = -e^{-d^2/2h^2}$ and generic (T, ε) no such pair (a, w) exists: the partition then depends on the sample through the basins of the density modes, which are not determined by the pairwise distances weighted independently of the assignment. [conjectural; not tested here]

Whatever the answer, the dynamic route is already occupied by mean shift, so a mechanical proposal of this kind has to be compared with mean shift, not with k -means, and the comparison must involve the trajectory, not the energy minimum.

5 Computational confirmation

Data and potentials. Five datasets: blobs, two moons and two circles from scikit-learn’s generators (300 points each in $\mathbb{R}^2$; `make_blobs` with three centres and std 0.9, $k = 3$; `make_moons`

potential	neg. type (Prop. 3.3)	partitions	max rel. error	min $\lambda_{\min}(\tilde{K})/\lambda_{\max}(\tilde{K})$	PSD
d^2 (Hooke)	yes	1015	3.6×10^{-12}	-2.6×10^{-16}	5/5
d^2 on the lift ($x, x ^2/s$)	yes	1015	2.6×10^{-12}	-3.2×10^{-16}	5/5
$(d-r)^2$ (rest length)	no	1015	1.1×10^{-11}	-5.5×10^0	0/5
d (constant tension)	yes	1015	2.2×10^{-13}	-4.7×10^{-18}	5/5
$1 - e^{-d^2/2\ell^2}$ (saturating)	yes	1015	1.2×10^{-13}	≥ 0	5/5
$\log(1 + d^2/\ell^2)$ (soft)	yes	1015	1.4×10^{-13}	≥ 0	5/5

Table 1: E1. Per potential, over the five datasets: whether Proposition 3.3 labels the potential of negative type (a label, not a computation); number of partitions tested; worst relative discrepancy between the spring energy and the kernel objectives; most negative normalised eigenvalue of the universal kernel $\tilde{K}$ across datasets; number of datasets on which $\tilde{K}$ is numerically PSD.

with noise 0.06 and `make_circles` with factor 0.45 and noise 0.05, $k = 2$), a random subset of 300 digits from `load_digits` (64 features scaled to $[0, 1]$, $k = 10$) and the standardised iris data (150 points, 4 features, $k = 3$). The six potentials of Section 2, with scales set from the data before any run: ℓ is the median distance to the seventh nearest neighbour, r the tenth percentile of all pairwise distances, s the median of $\|x\|^2$. In all, $5 \times 6 = 30$ configurations. Seeds: 20260930 (E1, E2), 20260931 (E3) and 20260932 (E4).

E1: identical objective values. *Criterion (predefined):* for every tested partition and every potential, $|E_{\text{sp}} - (\frac{1}{2}J_{K'} + \text{const})| \leq 10^{-9} \max(1, |E_{\text{sp}}|)$ with the constant of Theorem 3.2, for the three representatives $K' \in \{-A, -A + \sigma I, \tilde{K}\}$; pass means 100%. The partitions are 200 uniformly random ones per configuration, the outputs of Lloyd and of the single-move relaxation on $\tilde{K}$ (minimally shifted when indefinite), and the ground-truth labels. *Result:* 19285/19285 checks passed, worst relative error 1.1×10^{-11} (the largest errors occur with the shifted representative, where σ is up to 4.4×10^4 and the identity involves cancellation); the Hooke energy equals the k -means SSE to 3.9×10^{-16} . On equal-size partitions $E_{\text{tot}} = (n/k)E_{\text{sp}}$ to 2.2×10^{-16} , while on random partitions the ratio $E_{\text{tot}}/E_{\text{sp}}$ ranges over $[29.7, 197.5]$: the normalisation is a genuine choice. Table 1 also reports the spectrum of $\tilde{K}$: for the 25 configurations with a potential of negative type it is PSD in 25 (numerically, $\lambda_{\min}/\lambda_{\max} \geq -3.2 \times 10^{-16}$), and for the rest-length potential it is indefinite in 5/5.

E2: same optimiser, same initialisation, same trajectory. *Criterion:* a brute-force spring relaxation that recomputes the total energy from scratch for every candidate move (it sees only springs) and the kernel implementation of Hartigan’s method via (3) (it sees only K), started from the same random labels with the same sweep order, must produce identical label vectors after every sweep and final energies equal to 10^{-9} relative, in all runs. Subsets of 120 points, three initialisations per configuration. *Result:* 90/90 identical trajectories (mean 4.9 sweeps, at most 11), worst final-energy discrepancy 1.4×10^{-14} ; the brute-force runs took 56 s in total against 0.6 s for the kernel runs. In addition, hub-anchored Hooke springs equilibrated alternately (Remark 3.4(b)) reproduced scikit-learn’s Lloyd k -means from the same initial centroids in 15/15 runs, with inertia equal to the spring energy to 4.3×10^{-16} .

E3: relaxation versus kernel k -means on the same energy. Five optimisers see the same kernel: *lloyd* (batch Lloyd on $\tilde{K}$, minimally shifted only when indefinite, 5 configurations), *lloyd+relax* (Lloyd, then single-move polish), *relax* (single-particle relaxation from the initial labels), *anneal* (Metropolis on the spring energy, 40 sweeps of geometric cooling from the median $|\Delta E|$ of random moves down by 10^{-3} , then relaxation [15]) and the control *lloyd (naive)* (Lloyd on $-A + \sigma I$, Proposition 3.6). Initialisations are shared: 10 random labelings and 10 kernel k -means++ seedings [1] per configuration; annealing uses the first 5 random ones. E^*

potential	configurations reaching E^* (of 5)				restarts reaching E^* (%)			Lloyd f.p.	best
	relax	lloyd+relax	lloyd	naive	relax	lloyd+relax	lloyd	1-move stable	ARI= 1
d^2 (Hooke)	4/5	5/5	3/5	1/5	54	57	31	46/100	4/5
d^2 on the lift ($x, x ^2/s$)	5/5	5/5	4/5	2/5	80	74	69	73/100	5/5
$(d-r)^2$ (rest length)	5/5	4/5	1/5	1/5	53	49	4	4/100	4/5
d (constant tension)	5/5	5/5	2/5	1/5	42	43	30	43/100	5/5
$1 - e^{-d^2/2\ell^2}$ (saturating)	4/5	2/5	2/5	0/5	4	2	2	12/100	2/5
$\log(1 + d^2/\ell^2)$ (soft)	5/5	4/5	2/5	1/5	37	34	17	26/100	4/5

Table 2: E3 by potential (five datasets each). “Reaching E^* ” means within 10^{-6} relative of the best energy found by any method. “Lloyd f.p. 1-move stable”: Lloyd fixed points that no single move improves (all relaxation fixed points are Lloyd fixed points, Proposition 3.5(ii); 100/100 per potential). “best ARI= 1”: configurations in which the best partitions of relax and lloyd+relax coincide.

is the best energy found by any method in a configuration. *Criteria (predefined)*: (C3a) every relaxation and annealing fixed point is Voronoi-stable for the kernel used (Proposition 3.5 predicts 100%); (C3b) the relaxation is called distinguishable from the kernel family only if its best-of-restarts energy differs from that of lloyd+relax by more than 10^{-6} relative, either sign being reported since it would be a statement about optimisers of one function; (C3c) the fraction of Lloyd fixed points that are not single-move stable is reported. *Disclosure*: a pilot run used $-A + \sigma I$ as the only Lloyd kernel and found it inert (Proposition 3.6); $\widetilde{K}$ then replaced it as the baseline, which favours the kernel side of the comparison, and the naive shift was kept as a control. C3a–C3c and their thresholds were written before the pilot and were not changed; every number reported comes from the final code. *Results* (Table 2, Figure 1; per configuration in `results/tables_relaxation.md`): all 600/600 relaxation and 150/150 annealing fixed points are Voronoi-stable (C3a). Under C3b, 6 of 30 configurations are distinguishable (5 in favour of the relaxation, 1 against; largest relative difference 7.4×10^{-3} ; the split is optimiser noise: the `-fast` run gives 3–3 with the same 6/30); whenever the two best energies agree the best partitions coincide (ARI = 1; 0 exceptions), and overall ARI = 1 in 24/30. The relaxation reaches E^* in 28 configurations, lloyd+relax in 25, Lloyd alone in 14 and the naive shift in 6 (annealing, with 5 restarts only, is not comparable on this count: 20); the median relative excess over restarts (median across configurations) is 8.4×10^{-4} , 7.7×10^{-4} , 3.7×10^{-3} and 5.9×10^{-1} . Only 34% of Lloyd fixed points on the canonical kernel are single-move stable, 6% on the naive shift (C3c); the single-move methods are single-move stable in 600/600, 600/600 and 150/150 runs (measured; expected by construction). The best energies were recomputed spring by spring, worst discrepancy 4.3×10^{-13} . E3 is descriptive (counts over 30 configurations, modest restart budgets); no inferential claim is made.

Reading of E1–E3 for the research line. If the screened formulations were of the pairwise kind of Definition 2.1, the screening could not have found a mechanical effect: the spring energies *are* the kernel objectives (E1), the mechanical implementation of the relaxation *is* a kernel implementation (E2), and every relaxed state *is* a kernel k -means fixed point (E3). Differences between optimisers of the same function are visible (Lloyd alone reaches E^* less often than the single-move methods; the naive shift is worse still), and they are the only differences there are.

6 What can be claimed

Table 3 collects the mathematical claims with their status; it separates what closes the screening by proof from what was only measured. The measured items are those of Section 5 (E1: 19285/19285 identities, $\widetilde{K}$ PSD in 25/25 negative-type and indefinite in 5/5 rest-length configurations; E2: 90/90 identical trajectories, 15/15 anchored runs equal to Lloyd; E3: 600/600 and 150/150 Voronoi-stable fixed points, 6/30 configurations with different best energies, naive-shift Lloyd

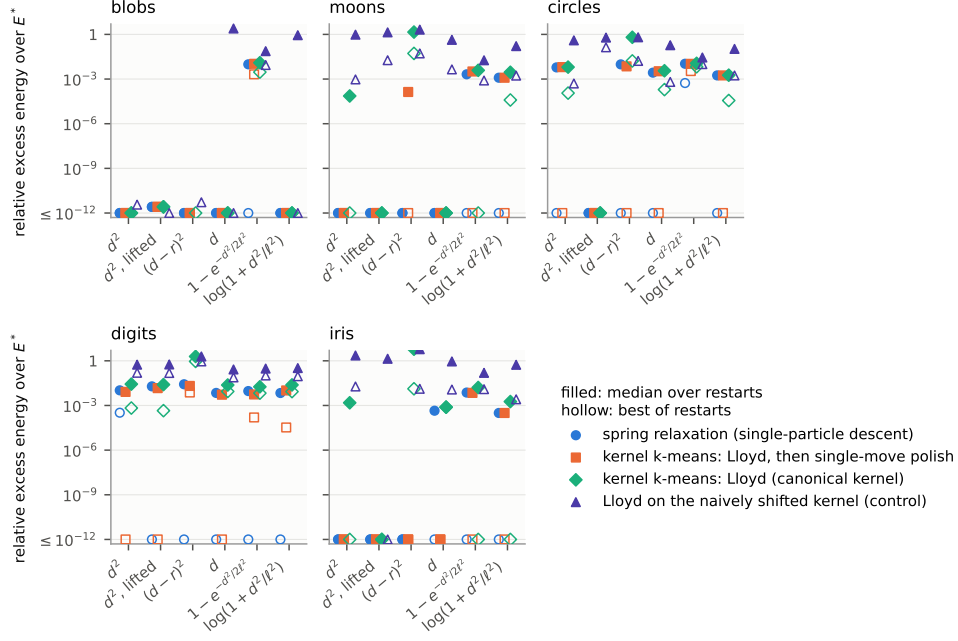


Figure 1: E3. Relative excess energy over E^* for the four optimisers of the same spring energy, per dataset and potential (filled: median over restarts; hollow: best of restarts; values below 10^{-12} are drawn at the floor). The optimisers differ in how often they reach the best local optimum, not in what they optimise; the naively shifted Lloyd is the control of Proposition 3.6.

at E^* in 6/30 against 14); “verified” means reproduced by the scripts of Appendix A with the fixed seeds, not proved.

Limitations. The family of Section 2 is a reconstruction from the line’s public summary; if the original screening used many-body terms, assignment-dependent parameters or a dynamics on positions, Section 4 applies and the theorem does not. The theorem is about objectives; it does not say that all optimisers of a spring energy behave alike, and E3 shows that they do not. The normalisation family is $\{1, 1/n_c\}$; other size weights (for instance the per-spring average) leave the exact identity and we only certify that they are not pairwise under the two standard weights. Theorem 4.1 certifies three specific functionals on two configurations, which is what “not in general” statements need, but says nothing about classes of terms or about how far from pairwise those energies are on typical data. The Bernstein composition result is cited, not re-proved in full. E3 uses one implementation of each optimiser with modest restart budgets; its numbers describe optimiser luck on five datasets, not a benchmark. Nothing is claimed about the dynamic route beyond its formulation.

Next steps for the research line. (a) Confirm with the author of the line that the screened formulations were of the pairwise form of Definition 2.1; if not, the emphasis moves to Section 4. (b) Settle Open problem 4.2 at least for the Gaussian flow, either by an explicit configuration where $P_{T,\varepsilon}$ is not a pairwise minimiser (the exact-arithmetic method of Theorem 4.1 applies once $P_{T,\varepsilon}$ is computed), or by a representation. (c) Characterise the size weights w for which $\sum_c w(n_c) \sum_{i < j \in C_c} A_{ij}$ is a weighted kernel k -means objective in the sense of Dhillon et al. [8]; the answer decides whether the normalisation is a mechanical degree of freedom or a graph-cut one. (d) If (a) is confirmed, the line’s summary can be updated from “no mechanical component survived” to “no mechanical component of the pairwise kind can survive”, with the open ingredients named.

claim	status
J is invariant under diagonal shift (up to $\sigma(n - k)$), constant and separable terms (Lemma 3.1)	proved here (elementary)
Per-particle spring energy = $\frac{1}{2}$ ratio association $-\frac{1}{2}k\phi(0) = \frac{1}{2}J_{-A} + \text{const}$, with a PSD representative after a shift (Thm. 3.2(b))	literature (ratio association $\leftrightarrow$ kernel k -means, 6, 8), stated here for spring matrices with the explicit constant
Zero-rest-length Hooke energy = k -means SSE (Thm. 3.2(c))	classical
Total energy / released tension = min-sum k -clustering / max- k -cut with weights A (Thm. 3.2(d))	proved here (elementary)
Every kernel k -means objective is a Hooke energy of a lifted configuration (Thm. 3.2(e))	classical (feature-space k -means)
Negative-type springs have a dataset-independent PD kernel and are Hooke springs after a lift and a $1/\sqrt{2}$ rescaling (Thm. 3.2(f))	proved here from Berg et al. [2]
d^2 is of negative type; Bernstein transforms preserve negative type (Prop. 3.3)	classical / literature
Rest-length springs: covered dataset-wise; no universal kernel exhibited ($\tilde{K}$ indefinite on all 5 datasets)	proved / open (verified numerically)
Lifting only changes the kernel; anchored Hooke springs are Lloyd k -means (Remark 3.4)	trivial / classical
Single-particle relaxation is Hartigan’s method; its fixed points are Lloyd fixed points for every PSD representative (Prop. 3.5)	proved here (generalises 27)
Lloyd’s rule depends on the PSD representative; large shifts freeze it (Prop. 3.6)	proved here; explicit form of an effect noted by Dhillon et al. [7, 8]
The cluster-mean rest length, the triangle-area three-body term, the per-spring average and the cross-normalisation tested are not pairwise objectives (Thm. 4.1)	proved here (exact arithmetic, two explicit configurations)
Dynamic (position-relaxation) mechanical clustering is not a pairwise objective (Conj. 4.3)	conjectural
Any statement about clustering quality on real data	not addressed

Table 3: Mathematical claims and their status.

A Reproducibility

The directory contains `experiments/mechanical.py` (potentials, energies, kernel forms, optimisers), `experiments/identity_check.py` (E1, E2, E4), `experiments/relaxation.py` (E3), `experiments/make_figures.py` and `experiments/make_numbers.py`, which converts `results/*.json` into the macros and table bodies used here, so that every number in the text is generated; `results/tables_identity.md` and `results/tables_relaxation.md` hold the full tables (E4 on both configurations; E3 per configuration). Both experiment scripts take `-fast`. The reference runs used seeds 20260930 (E1, E2; E4 uses its own generator with seed 20260932, so its configurations do not depend on `-fast`) and 20260931 (E3), Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1 [28] and scikit-learn 1.9.1 [18]; E1, E2 and E4 took 159 s and E3 128 s on shared cores. *Energies and tests.* E_{sp} and E_{tot} are computed from the definition (1) by summing ϕ over the within-cluster pairs of the distance matrix; J_K from (2). Partitions always have k non-empty clusters; a Lloyd step that would empty a cluster refills it with the worst-fitted point. Relaxation sweeps in a fresh random order; a particle moves to the argmin cluster only if the energy decreases by more than 10^{-10} times the initial energy scale, identically in both implementations of E2; a particle alone in its cluster does not move. Annealing proposes a uniformly random other cluster and accepts with probability $\min(1, e^{-\Delta E/T})$. Voronoi stability: own δ within 10^{-9} (relative to the largest $|\delta|$) of the minimum over clusters; single-move stability: no $\Delta E < -10^{-9}$ times the energy scale. E4. Integer coordinates drawn uniformly in $\{0, \dots, 10\}^2$ (repeated until distinct) from `default_rng(20260932)`; the design matrix has one column per pair and one for the constant; the normal equations are solved by exact Gaussian elimination over $\mathbb{Q}$.

References

- [1] D. Arthur and S. Vassilvitskii. k -means++: the advantages of careful seeding. In *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1027–1035, 2007.
- [2] C. Berg, J. P. R. Christensen, and P. Ressel. *Harmonic Analysis on Semigroups: Theory of Positive Definite and Related Functions*, volume 100 of *Graduate Texts in Mathematics*. Springer, New York, 1984.
- [3] M. Blatt, S. Wiseman, and E. Domany. Superparamagnetic clustering of data. *Physical Review Letters*, 76(18):3251–3254, 1996.
- [4] Y. Cheng. Mean shift, mode seeking, and clustering. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 17(8):790–799, 1995.
- [5] D. Comaniciu and P. Meer. Mean shift: a robust approach toward feature space analysis. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 24(5):603–619, 2002.
- [6] I. S. Dhillon, Y. Guan, and B. Kulis. Kernel k -means: spectral clustering and normalized cuts. In *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '04)*, pages 551–556. ACM, 2004.
- [7] I. S. Dhillon, Y. Guan, and B. Kulis. A unified view of kernel k -means, spectral clustering and graph cuts. Technical Report TR-04-25, Department of Computer Sciences, University of Texas at Austin, 2004.
- [8] I. S. Dhillon, Y. Guan, and B. Kulis. Weighted graph cuts without eigenvectors: a multilevel approach. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 29(11):1944–1957, 2007.
- [9] R. Durbin and D. Willshaw. An analogue approach to the travelling salesman problem using an elastic net method. *Nature*, 326:689–691, 1987.
- [10] K. Fukunaga and L. D. Hostetler. The estimation of the gradient of a density function, with applications in pattern recognition. *IEEE Transactions on Information Theory*, 21(1):32–40, 1975.
- [11] M. Girolami. Mercer kernel-based clustering in feature space. *IEEE Transactions on Neural Networks*, 13(3):780–784, 2002.
- [12] J. A. Hartigan. *Clustering Algorithms*. Wiley, New York, 1975.
- [13] J. A. Hartigan and M. A. Wong. Algorithm AS 136: A k -means clustering algorithm. *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, 28(1):100–108, 1979.
- [14] D. Horn and A. Gottlieb. Algorithm for data clustering in pattern recognition problems based on quantum mechanics. *Physical Review Letters*, 88(1):018702, 2002.
- [15] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi. Optimization by simulated annealing. *Science*, 220(4598):671–680, 1983.
- [16] S. P. Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982.
- [17] J. MacQueen. Some methods for classification and analysis of multivariate observations. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, pages 281–297. University of California Press, 1967.
- [18] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [19] K. Rose. Deterministic annealing for clustering, compression, classification, regression, and related optimization problems. *Proceedings of the IEEE*, 86(11):2210–2239, 1998.
- [20] K. Rose, E. Gurewitz, and G. C. Fox. Statistical mechanics and phase transitions in clustering. *Physical Review Letters*, 65(8):945–948, 1990.
- [21] S. Sahni and T. Gonzalez. P-complete approximation problems. *Journal of the ACM*, 23(3):555–565, 1976.
- [22] R. L. Schilling, R. Song, and Z. Vondraček. *Bernstein Functions: Theory and Applications*, volume 37 of *De Gruyter Studies in Mathematics*. De Gruyter, Berlin, 2nd edition, 2012.
- [23] I. J. Schoenberg. Remarks to Maurice Fréchet’s article “sur la définition axiomatique d’une classe d’espaces distanciés vectoriellement applicable sur l’espace de Hilbert”. *Annals of Mathematics*, 36(3):724–732, 1935.
- [24] I. J. Schoenberg. Metric spaces and positive definite functions. *Transactions of the American Mathematical Society*, 44(3):522–536, 1938.
- [25] B. Schölkopf, A. Smola, and K.-R. Müller. Nonlinear component analysis as a kernel eigenvalue problem. *Neural Computation*, 10(5):1299–1319, 1998.
- [26] S. Z. Selim and M. A. Ismail. K-means-type algorithms: a generalized convergence theorem and characterization of local optimality. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 6(1):81–87, 1984.
- [27] M. Telgarsky and A. Vattani. Hartigan’s method: k -means clustering without Voronoi. In *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 9 of *JMLR Workshop and Conference Proceedings*, pages 820–827, 2010.

- [28] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, I. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, and SciPy 1.0 Contributors. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020.
- [29] W. E. Wright. Gravitational clustering. *Pattern Recognition*, 9(3):151–166, 1977.
- [30] A. L. Yuille. Generalized deformable models, statistical physics, and matching problems. *Neural Computation*, 2(1):1–24, 1990.
- [31] H. Zha, X. He, C. Ding, M. Gu, and H. D. Simon. Spectral relaxation for k -means clustering. In *Advances in Neural Information Processing Systems 14 (NIPS 2001)*, pages 1057–1064. MIT Press, 2002.